

# Supplementary Material: CAdam

## S1 End-to-End Pseudocode

Algorithm S1 summarizes how CAdam is integrated into an optimization based generative 3DGS pipeline. The renderer, camera sampler, generative guidance loss, Adam optimizer, and the host training loop are unchanged. CAdam only replaces the density-control decisions: the densification candidate mask and the opacity-reset rule. Following the host 3DGS training order, density control is evaluated after backpropagation and before the current Adam optimizer step. Thus, at a scheduled density-control call, CAdam reads the Adam moment states accumulated from previous optimizer updates, while the current gradient is used by the subsequent Adam step.

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### Algorithm S1 End-to-End Pseudocode of CAdam.

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**Require:** Initial Gaussians, camera sampler, guidance loss, Adam optimizer

**Require:** Densification interval  $K_{\text{den}}$ , opacity-reset interval  $K_{\text{op}}$

**Require:** CAdam thresholds  $\tau_Q$ ,  $\tau_{\text{SNR}}$ , scale threshold  $\tau_{\text{scale}}$

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1: Initialize Gaussians, Adam states, and visibility-aware ages  $a_i$ 
2: for  $t = 1$  to  $T$  do
3:   Sample cameras and render images from Gaussians  $\mathcal{G}$ 
4:   Record the visible Gaussian set  $\mathcal{V}_t$ 
5:   Compute the generative guidance loss  $\mathcal{L}$  and backpropagate
6:   if  $t$  is a scheduled densification iteration then
7:     for all  $G_i \in \mathcal{G}$  do
8:       Read Adam moments  $(\mathbf{m}_i, \mathbf{v}_i)$  of the position  $\mu_{xyz,i}$ 
9:       Bias-correct  $(\mathbf{m}_i, \mathbf{v}_i)$  using the visibility-aware age  $a_i$ 
10:      Compute the momentum score  $s_i \leftarrow \|\hat{\mathbf{m}}_i\|_2$ 
11:      Compute  $\text{SNR}_i$  from  $(\hat{\mathbf{m}}_i, \hat{\mathbf{v}}_i)$  as defined in the main paper
12:    end for
13:     $q \leftarrow \text{Quantile}(\{s_i\}_{i=1}^{|\mathcal{G}|}, \tau_Q)$ 
14:     $\mathcal{M}_{\text{densify}} \leftarrow \{G_i \mid s_i > q \text{ and } \text{SNR}_i > \tau_{\text{SNR}}\}$ 
15:    Clone Gaussians in  $\mathcal{M}_{\text{densify}}$  with  $\|S_i\|_\infty \leq \tau_{\text{scale}}$ 
16:    Split Gaussians in  $\mathcal{M}_{\text{densify}}$  with  $\|S_i\|_\infty > \tau_{\text{scale}}$ 
17:    Remove split parents
18:    Zero-initialize Adam states, and ages for new Gaussians
19:    Prune Gaussians with low opacity or invalid scale
20:    Apply the same pruning mask to tensors, Adam states, and ages
21:  end if
22:  if  $t$  is a scheduled opacity-reset iteration after warm-up then
23:    for all  $G_i \in \mathcal{G}$  do
24:      Compute  $\text{SNR}_i$  from the Adam moments of  $\mu_{xyz,i}$ 
25:    end for
26:    Cap opacity at  $\alpha_{\text{reset}}$  only for Gaussians with  $\text{SNR}_i < \tau_{\text{SNR}}$ 
27:    Reinitialize Adam states of the modified opacity parameters
28:    Preserve Adam states of the world-space position parameters
29:  end if
30:  Perform the Adam optimizer step
31:  Increment  $a_i$  for  $G_i \in \mathcal{V}_t$  with a valid position-gradient contribution
32:  Clear gradients
33: end for

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## S2 Implementation Details

*Setup.* We implement CAdam as a density-control module for optimization-based generative 3D Gaussian Splatting. For the controlled experiments and ablations, we use a representative optimization based 3DGS pipeline with VFDS loss and Stable Diffusion 3.5 guidance, following the setup in the main paper. The renderer, camera sampling, guidance loss, optimizer, and training loop are kept identical between the baseline and CAdam. Only the densification mask and opacity-reset rule are replaced. Unless otherwise specified, each prompt is initialized with 100,000 Gaussians and optimized for 5,000 iterations with batch size 4 on a single NVIDIA RTX A5000 GPU with 24GB memory. All Gaussian parameters are optimized with Adam.

For the standard baseline, the host pipeline uses the usual finite-window positional-gradient accumulator with window size  $K = 100$ . CAdam keeps the same  $K_{\text{den}}$  step density-control schedule for fair comparison, but it does not use the scalar gradient accumulator itself. Instead, it reads the Adam moment states of the world-space Gaussian positions.

*Moment and SNR.* CAdam does not introduce a separate momentum estimator. It directly reuses Adam's first- and second-moment states for the world-space position parameter  $\mu_{xyz,i}$  of each Gaussian  $G_i$ . These states correspond to the world-space positional gradient  $\mathbf{g}_i = \nabla_{\mu_{xyz}} \mathcal{L}$  used in the main paper.

For bias correction, we maintain a visibility-aware age  $a_i$  for each Gaussian. The age is incremented once per optimizer step when  $G_i$  is visible in the minibatch and receives a valid position-gradient contribution. This per-Gaussian age is used instead of the global iteration index, which avoids over-correcting newly created or rarely visible Gaussians. For numerical stability, the Adam bias-correction denominator is clamped from below by  $10^{-3}$ .

At each scheduled densification call, we compute the momentum score  $s_i = \|\hat{\mathbf{m}}_i\|_2$  from the bias-corrected first moment. We also compute the Intrinsic SNR from  $\hat{\mathbf{m}}_i$  and  $\hat{\mathbf{v}}_i$ , using the same definition as in the main paper with  $\epsilon = 10^{-9}$ . A Gaussian passes the CAdam gate only when it satisfies both conditions: its momentum score is above the  $\tau_Q$  quantile of the current Gaussian population, and its Intrinsic SNR is larger than  $\tau_{\text{SNR}}$ . This implements the Context-Adaptive Selection rule: the quantile test measures relative importance within the current scene, while the SNR test verifies signal reliability.

*Adam-state management.* CAdam decouples candidate selection from the structural operation. Once a Gaussian passes the CAdam gate, the actual operation follows the standard scale rule from 3DGS. Selected Gaussians with  $\|S_i\|_\infty \leq \tau_{\text{scale}}$  are cloned, while selected Gaussians with  $\|S_i\|_\infty > \tau_{\text{scale}}$  are split. Thus, CAdam changes when a Gaussian is selected, but it does not introduce a new geometric split/clone criterion.

When a Gaussian is cloned, the newly created Gaussian copies the parent's geometric and appearance parameters. However, its Adam states and visibility-aware age are initialized to zero. When a Gaussian is split, the children are initialized according to the

host 3DGS split operation, the parent is removed, and the children also start with zero Adam states and zero age. This prevents the momentum history of a parent Gaussian from being treated as verified evidence for newly instantiated primitives.

After pruning, all Gaussian parameter tensors, Adam states, and visibility aware ages are filtered by the same pruning mask. This keeps the representation, optimizer state, and CAdam bookkeeping aligned after every topology update.

**Selective opacity reset.** CAdam replaces the global opacity reset with an SNR-aware selective reset. After the initial warm-up, opacity reset is evaluated at the host pipeline’s scheduled reset interval. Instead of resetting all Gaussians, CAdam only suppresses unreliable primitives whose Intrinsic SNR is below  $\tau_{\text{SNR}}$ .

In implementation, the reset is applied as an opacity cap with  $\alpha_{\text{reset}} = 0.01$ . For a low-SNR Gaussian, opacity values above this cap are set to  $\alpha_{\text{reset}}$ , while already lower opacity values are left unchanged. Gaussians that pass the SNR gate keep their opacity unchanged. After the reset, the Adam states associated with the modified opacity parameters are reinitialized. The Adam states of the world-space position parameters are preserved, because these states provide the momentum and SNR statistics used by CAdam.

**Scheduling and hyperparameters.** Unless otherwise specified, optimization runs for 5,000 iterations. Density control is evaluated every  $K_{\text{den}} = 100$  iterations for  $100 < t < 4,000$ , so scheduled densification calls occur from iteration 200 to 3,900. The opacity-reset interval is  $K_{\text{op}} = 400$ . After warm-up, selective opacity reset is triggered when  $t > 400$  and  $t$  is divisible by 400 while density control is active; the first reset therefore occurs at iteration 800.

The default CAdam thresholds are  $\tau_Q = 0.9$  and  $\tau_{\text{SNR}} = 0.1$ . Adam uses  $\beta_1 = 0.9$  and  $\beta_2 = 0.999$ . The SNR stabilizer is  $\epsilon = 10^{-9}$ . The opacity reset cap is  $\alpha_{\text{reset}} = 0.01$ . The scale threshold  $\tau_{\text{scale}}$  is set to 0.003 times the scene extent. We prune Gaussians with opacity below 0.01, maximum scale below  $10^{-8}$  times the scene extent, or maximum scale above 0.35 times the scene extent.

These hyperparameters are fixed across the main experiments unless otherwise noted. The threshold sensitivity analysis in the main paper varies only  $\tau_Q$  and  $\tau_{\text{SNR}}$  while keeping the remaining training schedule, optimizer settings, split/clone rule, and pruning rule unchanged.

### S3 Quantitative Ablation

To complement the qualitative ablation in the main paper, Table S1 reports a compact component-wise quantitative ablation. We use the same prompt as the qualitative ablation in the main paper: “*light-house, full view, smooth white cylindrical tower, detailed red lantern room, HDR, photorealistic, 8K.*” This single-prompt evaluation is intended as a controlled diagnostic to align with the qualitative ablation setting in the main paper, rather than as an aggregate benchmark across prompts. All variants are evaluated under the same controlled setup described in Sec. S2. The ablation is cumulative: **Baseline** uses standard view-space magnitude-based densification, **+Moment** replaces the scalar gradient accumulator with the Adam first-moment score, **+Context-Adaptive Selection** further applies the quantile and Intrinsic SNR gates, and **+Selective Opacity Reset**

corresponds to the full CAdam configuration. The **Baseline** uses the host pipeline’s default densification threshold  $\tau_{\text{pos}} = 0.00075$ . Since **+Moment** changes the selection signal from view-space accumulated magnitude to the Adam first-moment score, its fixed threshold is not directly comparable in scale; we use an empirically calibrated threshold  $s_i > 0.0075$  and keep it fixed for this diagnostic. We report the final number of Gaussians, ImageReward, and HPS v2, covering representation efficiency, semantic alignment, and perceptual aesthetics in a compact form.

Table S1. Component-wise quantitative ablation of CAdam. Baseline uses the host pipeline’s default view-space densification threshold  $\tau_{\text{pos}} = 0.00075$ ; **+Moment** uses the fixed momentum-score threshold  $s_i > 0.0075$ ; adaptive variants use  $\tau_Q = 0.9$  and  $\tau_{\text{SNR}} = 0.1$ . We report the final number of Gaussians, ImageReward, and HPS v2 under the same training schedule, initialization protocol and guidance model.

| Variant                         | # Gaussians ↓ | ImgRwd ↑ | HPS v2 ↑ |
|---------------------------------|---------------|----------|----------|
| Baseline                        | 5853K         | -0.475   | 0.247    |
| +Moment                         | 2572K         | -0.559   | 0.257    |
| +Context-Adaptive Selection     | 31K           | -1.457   | 0.119    |
| +Selective Opacity Reset (Full) | 75K           | -0.397   | 0.248    |

### S4 Depth and Multi-view Visualization

To complement the perceptual and text-alignment metrics in the main paper, Fig. S1 provides geometry-oriented visualizations. For each example, we render RGB images from multiple viewpoints and the corresponding depth maps using the final optimized Gaussian representation. The depth maps are normalized per object for visualization. These results should be interpreted as complementary evidence: CAdam improves density-control stability, but it does not replace explicit 3D supervision or a multi-view-consistent generative prior.

### S5 Failure Examples

Fig. S2 shows representative failure cases. CAdam targets density control in optimization-based generative distillation, and therefore does not impose global semantic or multi-view consistency on the underlying generative guidance. As a result, when the guidance model provides view-dependent or semantically ambiguous pseudo-targets, the optimized representation may still exhibit Janus-like artifacts, inconsistent side/back views, or incomplete object structure. These examples highlight the remaining limitations of CAdam’s scope: it can suppress redundant Gaussian growth and stabilize densification, but it does not resolve semantic or multi-view inconsistency inherited from the guidance model.

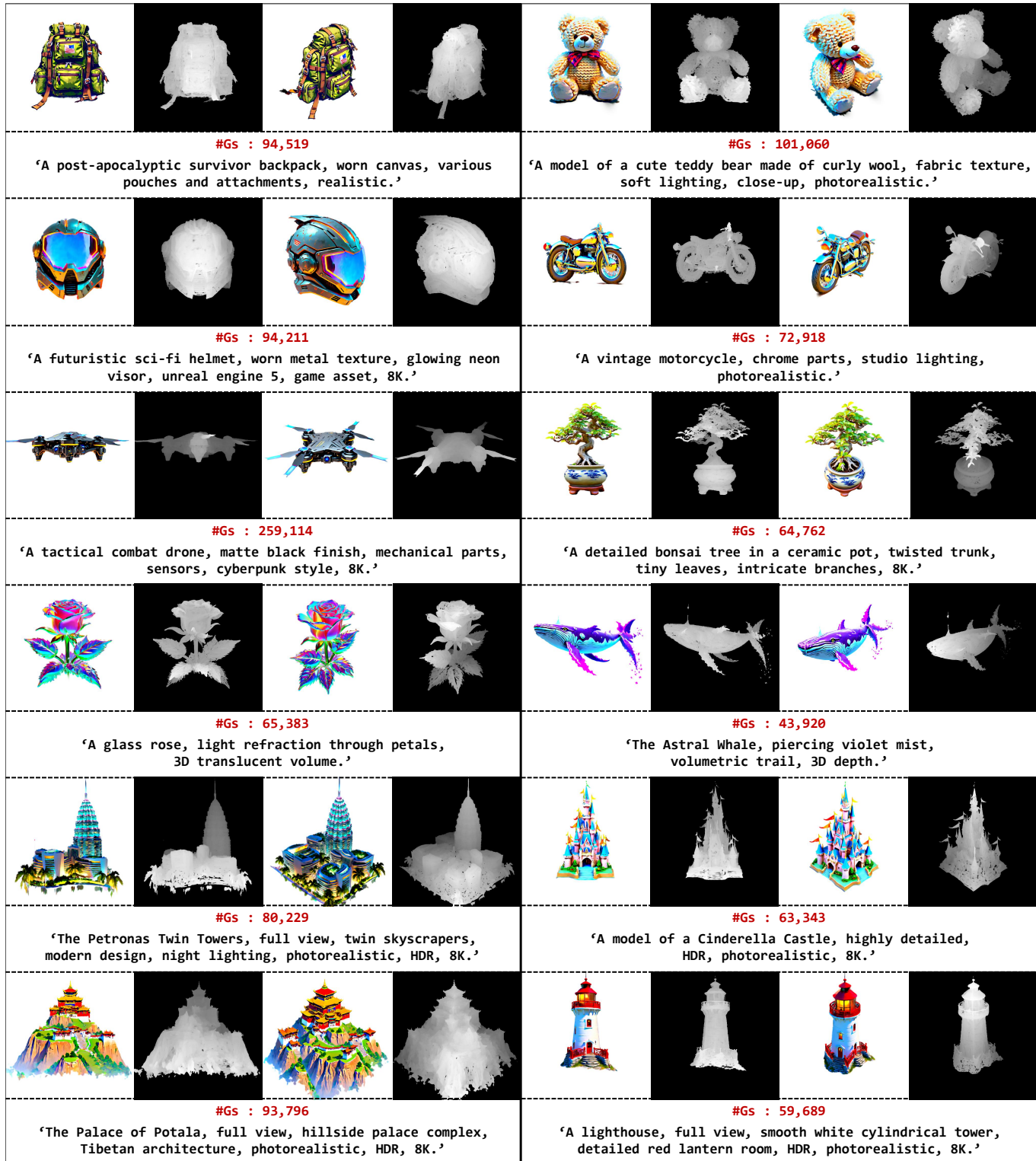


Fig. S1. Depth and multi-view visualizations of representative CAdam results. Each example shows RGB renderings from multiple viewpoints together with depth maps rendered from the same optimized Gaussian representation.

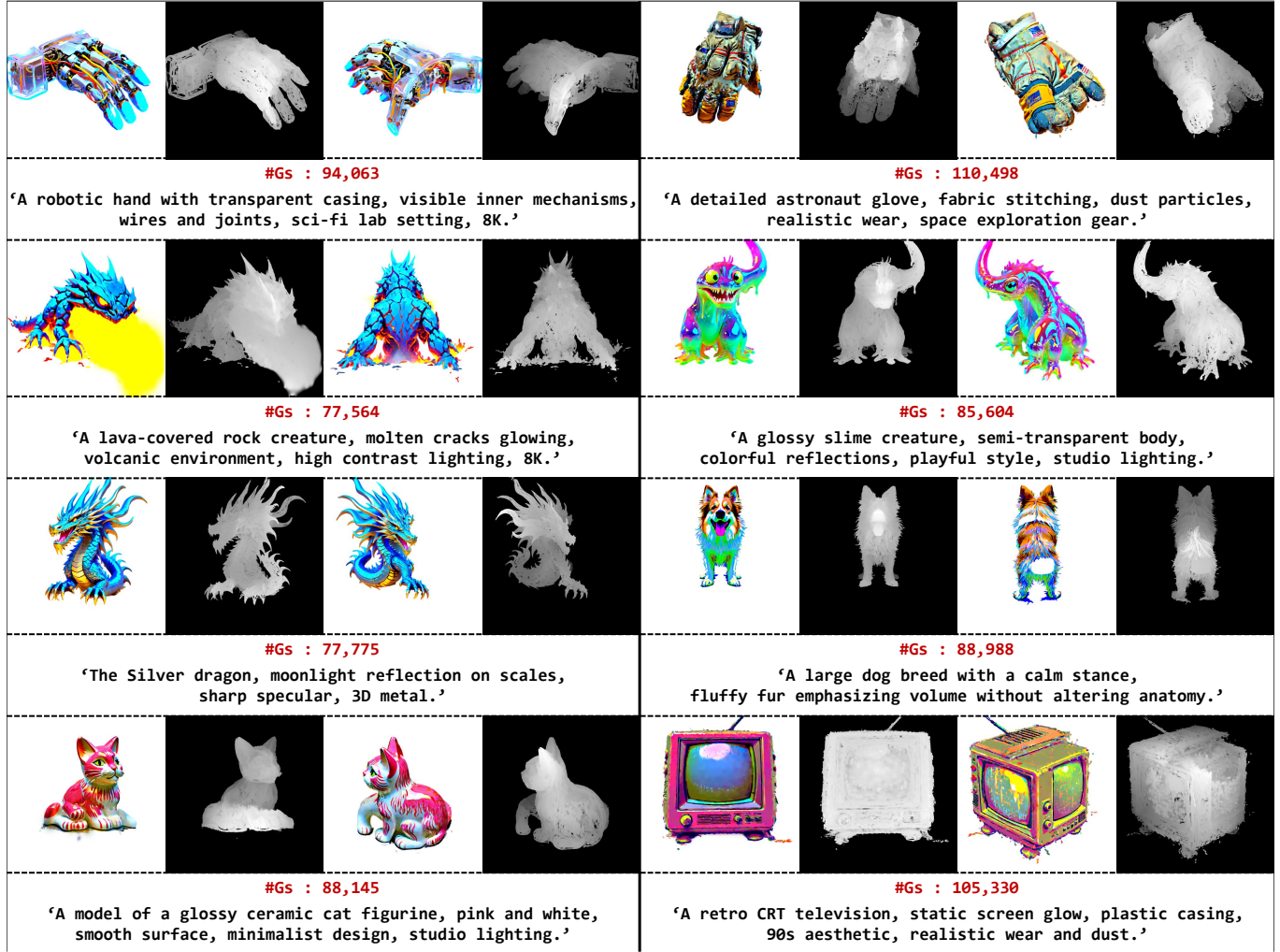


Fig. S2. Failure examples. CAdam improves density control but does not enforce global semantic or multi-view consistency. For challenging prompts, the underlying guidance model can still produce view-dependent or semantically ambiguous pseudo-targets, leading to Janus-like artifacts, inconsistent side/back views, or incomplete structures.